

1. Executive Summary

Project title: Are the Elderly Populations in Florida Going to Drown?: A GIS Investigation

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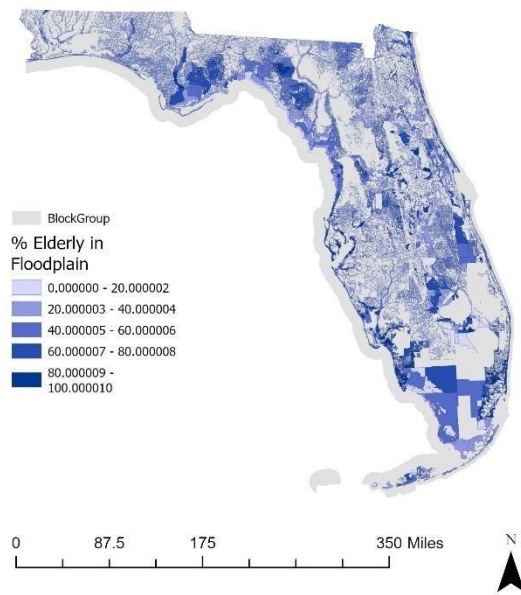
SUMMARY OF PROJECT:

Understanding and visualizing where vulnerable populations who live within the flood plains of Florida can play a crucial role in emergency preparedness, resource allocation, and safety regulations. Our analysis focuses on the elderly residents (age 65+) of Florida who are one of many examples of vulnerable individuals when it comes to the damaging effects of flooding because they often require additional assistance during instances of emergency, including mobility and transportation support and medical accommodations, which could potentially lengthen the time it takes to follow evacuation procedures. To visually identify and gather statistical data for these elderly populations, the Florida Office of Floodplain Management (OFM) can be more precise and accurate when it comes to improving their emergency preparedness and procedures.

For this analysis, we conducted Regression and Spatial Autocorrelation - Hot Spot Analyses. From the Regression results we searched for data which would indicate demographic factors such as population density, racial and ethnic diversities, proportion of those under the poverty line, etc. to observe whether these factors can provide statistically significant evidence which shape the distribution of the elderly population among the flood zones. This data found that only ~3% of the proportion of elderly can be explained by the variables. Following these findings, we observed within our Spatial Autocorrelation analysis that there is evidence of strong clustering based off our Moran's Index of 0.103 and a highly significant z-score of 317.81 ($p < 0.0001$). These results show that the elderly populations are not randomly distributed and are in high-high clusters, specifically around the coastal areas, in the panhandle, and in most of the southern half of Florida.

These findings can be significant for the OFM as they highlight where the elderly populations are clustered in the most impacted areas of Florida's flood plains, thus why they should be taken seriously and prioritized when thinking about emergency preparedness and action. As a next step, integrating these results into the operation plans could help to improve evacuation routes, resource allocations, and mitigation overall which can help to strengthen the state of Florida's ability to support its most vulnerable residents such as the elderly population.

Flood Zones
Source: FGDL & Census



Population Proportion Age 65+ : Total Population
Source: Census

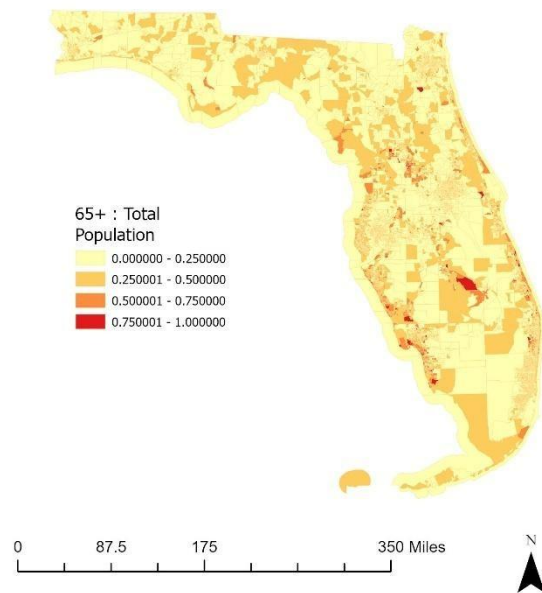


Figure 1: Flood Zones (Left): Represents the spatial distribution of flood plains across the state of Florida. Proportion of residents 65+ in proportion to the Total Population of Florida (Right): Represents this ratio by block group. Data sources: FGDL and U.S. Census Bureau.

2. Results

Method 1: Spatial Autocorrelation

Statistic	Value
Moran's Index	0.102665
Expected Index	−0.000075
Variance	0.0001
z-score	317.808788
p-value	< 0.0001*

Table 1. Global Morans I, all significant *alpha < 0.01

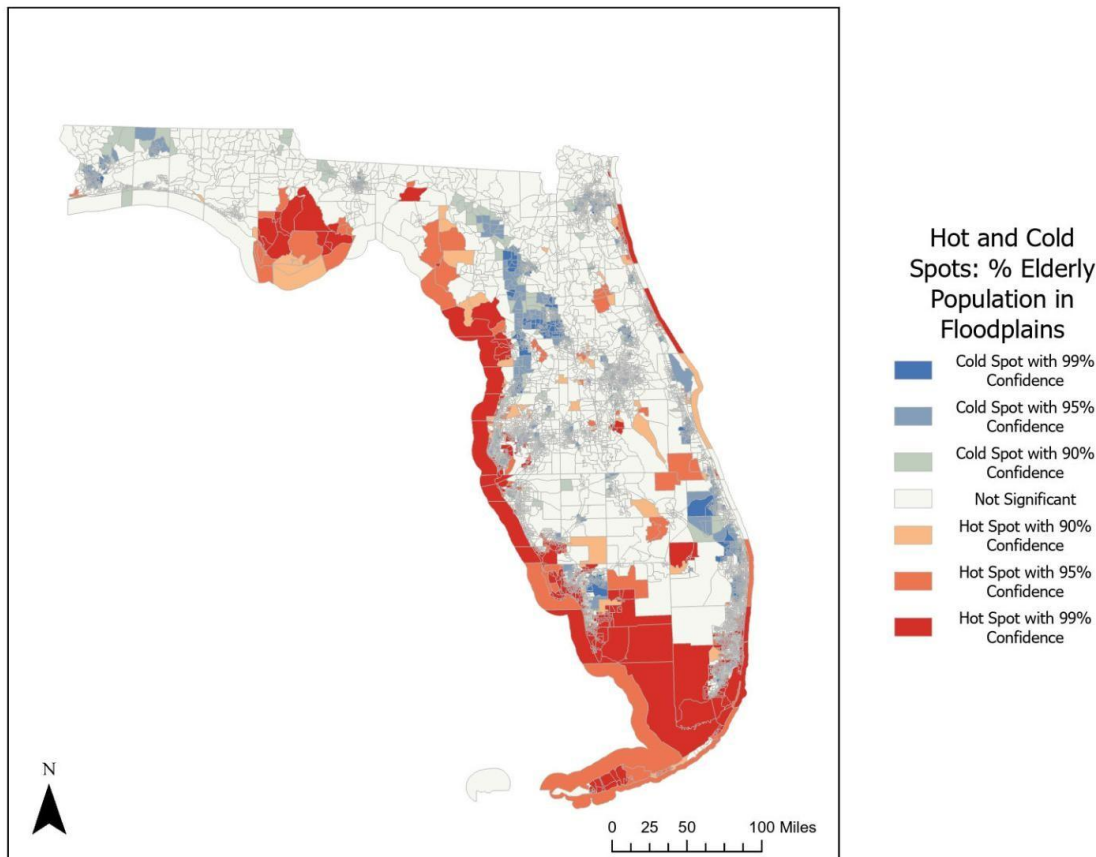


Figure 2: Represents a Hot and Cold spot analysis of the proportion of elderly residents living within Florida’s floodplains. Created using Getis-Ord Gi* spatial autocorrelation tool in ArcGIS Pro v3.6.0. Red areas indicate values of high cluster significant at 99%, 95% and 90%, while blue areas represent significant low-value clusters at the same percentage.

Global Morans I results

Morans I: a positive result of (0.103) indicates that the block groups have similar values which tend to cluster around one another.

Variance: The variance reflects how much the Morans I would change if this pattern was random. Since we have a very low variance value of 0.0001, this is indicative that there is clustering and this indeed is not random.

Z Score: With a large z-score of 317.809, we can without a doubt see that there is evidence of clustering and this pattern was not formed by random chance. This can be further backed up by the Hot and Cold spot map which presents areas of high spatial autocorrelation of the proportion of elderly living within the flood plain.

P-Value: The p-value of (0.0001) explains the probability that our analysis was caused by random chance. From this, we can infer that this pattern is again not random and is clustered considering how low our p-value is.

Getis-Ord Gi* Hot Spot analysis Results

There are distinct hot and cold spots which explain the proportion of elderly living within the floodplains.

The hot spots are mainly within the coastal regions of the state. I will point out that some of the hot spots are also within the ocean blocks of the flood plains. Due to this, some of the clusters around the coast might be calculated higher to what they actually are; however, this is not to say it is wrong, these clusters are still important and provide relative information as to where the proportion of elderly are clustered within and around the flood plains. This can also be backed by our “65+ population map” which identifies the density of elderly residents in the state of Florida.

The 99-90% clusters in the coastal region of Florida are primarily in South Florida, the panhandle, as well as on the eastern side of the state. These areas signify where high proportions of elderly residents live within these flood plains, and where the block groups of similar values share these high proportions, thus why we see significant high-high clusters.

Cold spots are in the inland/ interior region of Florida, which isn't to say no elderly populations are in these areas, but is to say that they have few elderly residents in those flood regions and do not cluster as highly.

Using the Hot Spot findings from this Spatial Autocorrelation Analysis, Florida Stakeholders such as Jillian Kraynak at the OFM would be able to identify where elderly populations are disproportionately concentrated in the high floodplain regions. By using this data, the OFM could be more informed when creating evacuation plans and response routes in cases of flooding events. Thus, they understand where to better allocate funding and resources. Alternatively, the cold spot findings from this analysis can be seen as areas which do not require as much funding or resources which can help prevent misallocation of funds which would in turn produce more accurate and enhanced emergency preparedness procedures.

Method 2: Multivariate Regression

Variable	Coefficient	Probability
Constant	0.204672	< 0.001*
Population Density	0.000006	< 0.001*
Proportion of Population Non-White	0.135122	< 0.001*
Proportion of Population Below Poverty Level	-0.132177	< 0.001*
Number of Observations	13388	
R-Squared	0.030464	
Adjusted R-Squared	0.030247	
F-Statistic	140.181027	< 0.001 *

Table 2. Multivariate Regression Results, all significant *alpha < 0.01

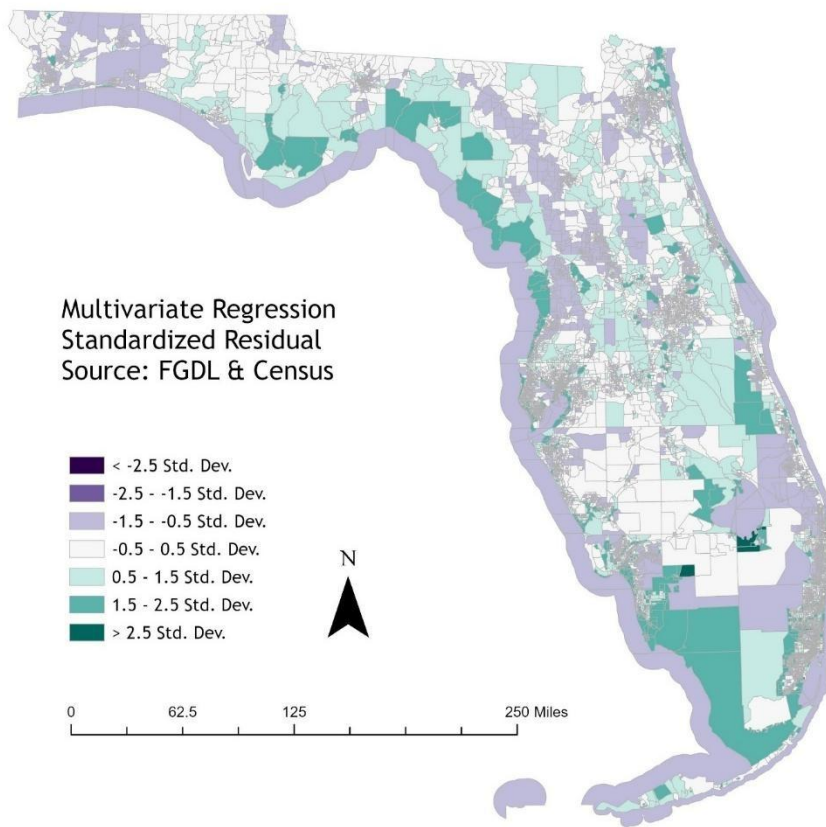


Figure 3. Residuals from Multivariate Regression. Green coastal areas reveal under-prediction, while purple pockets reveal over-prediction

The multivariate regression helps explain some of the patterns seen in the clustering of elderly populations in the floodplain. All three variables were significant at the 0.01 level, so the relationships we're seeing are valid. Population density has a small positive coefficient (0.000006), meaning that in more built-up areas there is a slight increase in the proportion of elderly living in a floodplain. The proportion of non-white residents also has a positive coefficient (0.135122), so places with more non-white populations tend to have more elderly living in these flood-risk areas. In contrast, the poverty coefficient (-0.132177) shows that block groups with higher poverty levels tend to have fewer elderly in the floodplain.

The model overall is significant (< 0.001), but the adjusted R^2 is only 0.030247. This means that while these variables do explain some of the variation, it's not a lot. Only about 3%

of elderly exposure to floodplains can be accounted for by population density, population nonwhite, and population below the poverty line. So even though they all matter statistically, they don't tell the full story of why these patterns look the way they do across Florida.

These results line up with what is seen in Figure 2. There's clear clustering of elderly residents in and around floodplains, but the demographic variables we tested do not fully explain why those clusters show up where they do. This points to other factors that are probably playing a much bigger role, like coastal development patterns, where retirement communities were built, local elevation, and how neighborhoods have grown over time. Because we wanted to understand if older adults live in floodplains for any identifiable reason, these weak relationships suggest that it's not driven by one simple demographic factor. Instead, it is likely a combination of geographic and development conditions that are not captured in basic correlations. In the future, a more refined multivariate model is needed to see if the combination of other factors together helps explain the pattern more clearly.

For stakeholders like Jillian Krynak at the OFM, our findings support their ongoing work to modernize flood maps and improve risk communication. Since elderly exposure to floodplains appears in specific clustered areas, these results can help OFM prioritize which regions may need updated mapping and targeted outreach. This is especially important for senior communities that may rely more on printed materials or local communication rather than digital platforms.

Finding	Evidence		(Map/Table Reference)	Why it Matters (research/ stakeholder)
Variable	Coefficient	Probability	Regression	
Constant	0.204672	< 0.001*	Regression	It shows the baseline level of the dependent variable. It helps OFM justify statewide minimum standards instead of

				only targeting the high-density communities.
Population Density	0.000006	< 0.001*	Regression	Positive coefficient means a higher density; this means that there is a higher flood related outcome. These areas often include senior heavy communities that can guide the OFM towards a grant application to FEMA.
Proportion of Population Non-White	0.135122	< 0.001*	Regression	This shows that nonwhite communities experience higher flood impacts or more exposure to them. The OFM is legally required to consider equity under the federal funding rules. It can also help them justify equitable outreach.
Proportion of Population Below Poverty Level	-0.132177	< 0.001*	Regression	This shows that poorer areas show lower values. That can mean for the OFM that poor neighborhoods are in less flood prone areas, or there is low insurance coverage for those areas. For the OFM, it shows that seniors in poverty have less of an ability to evacuate or recover financially and can't be overlooked.

R-Squared	0.030464	-	Regression	This shows that flood impacts are shown by many factors and not only demographics. For the OFM this can justify
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				more data collection for better modeling and that any policy must consider more than just population characteristics.
Adjusted R-Squared	0.030247	-	Regression	This shows that the model isn't overfitting and this is good for the OFM because this can help ensure the data is based on reliable statistical relationships and not randomness.
F-Statistic	140.18102	< 0.001 *	Regression	This shows that this is statistically significant data. For the OFM it shows them that the variables used are meaningful for predicting flood risks in the area. It also strengthens arguments for policy changes.
Moran's Index	0.102665	-	Spatial Autocorrelation	It shows that locations with similar values are clustered together for the OFM this shows that it isn't random and helps prioritize regions for seniors.

Expected Index	-0.000075	-	Spatial Autocorrelation	This is important to the OFM because they must use spatially informed planning and not a generic state wide approach.
Variance	0.0001	-	Spatial Autocorrelation	This shows the uncertainty of the autocorrelation estimate. Since it is low variance then there is high reliability. This matters for the OFM because they must
				defend the decisions made when trying to apply for funding.
z-score	317.80878	-	Spatial Autocorrelation	This shows that the high z-score equals a very strong cluster. This can help justify a need for regional scale projects and can support targeting senior citizens in the highrisk communities
p-value	< 0.001*	-	Spatial Autocorrelation	This shows the spatial clustering is not due to chance the OFM needs this to show the reliability when going to FEMA for risk assessments.

Table 3. Summary of Results *alpha < 0.01

3. Methods Reflection

In our study, we agreed that the spatial autocorrelation analysis yielded the best insights for our research investigation. Although our regression model provided us with some significant results, only ~3% of the variation in the proportion of elderly residents living within the floodplain could be explained by our variables of population density, racial and ethnic diversity, and poverty levels.

In contrast, the Spatial Autocorrelation analysis showed a visual of the geographic clustering which allowed us to see exactly where the hot and cold spots were across Florida and if they were clustered, dispersed, or truly randomly distributed.

Floodplain data is difficult to download. At first, FEMA was used to download data, we were planning on using a smaller scope for our project, possibly only one count. After evaluation, we decided to do the entire state, but data had to be downloaded by each county. It was also decided to use a smaller scale scope to capture as much detail as possible. We decided to use block groups. After searching, data was found from the Florida Geographic Data Library to download for the entire state.

For the spatial autocorrelation, a fixed distance matrix for neighbors was originally picked. This yielded one large hot spot and one large cold spot. After changing the matrix to k nearest neighbor, there was a meaningful outcome.

With regression, originally used an independent variable as % of block groups in flood plain which led to a r^2 value of 96%. This % of elderly in floodplain were calculated as % in block group so that would explain the large influence. After this, % of block group in floodplain was removed so the independent variables remained population density, proportion of non-white population, and proportion of population below poverty level.

If we were to repeat this project, the main thing we would do is further investigate environmental factors like elevation or hydrogeological features which could increase the

chance of flooding more in one area compared to another. Along with this, we think it would have been a neat idea to evaluate the current evacuation route plan the state of Florida has, thus so we could improve it in some way.

Our data is from credible government sources, and up to date within the past five years and appropriate for studying floodplain exposure and where the elderly population lives. For the regression model, it did not explain as much variation as expected; even with significant coefficient, the model only captured a small portion of why elderly appear where they do, which shows other key variables were not included in our data.

The main limitations were the variables chosen to analyze why elderly are in a floodplain and focused only on demographics while there should have been incorporation of more development-based predictions such as elevation differences or retirement-oriented planning. The hotspots results include coastal blocks that fall directly into the water so that could inflate clustering near the shoreline.

Finally, we also relied on block-group–level data, which smooths out variation within communities. Elderly residents might concentrate in specific neighborhoods, apartment complexes, or retirement communities that don't show up clearly on the block group scale. This can make the risk appear more widespread than it is. Also, the model used did not include elderly population near a floodplain, only populations IN a floodplain. Elderly people may live next to a floodplain and still face similar levels of risk but not captured in our dependent variable. ---

4. Stakeholder Integration

Jillian Kraynak is likely prioritizing ways to reduce flooding risk through mapping and technical ways. The OFM's mission is to promote floodplains and improve flood hazard mapping so communities can reduce risk to the community. By doing our research it goes hand in hand with what the OFM's mission is and can help become more accurate for the senior citizen population and people in the community, to help minimize the risks of flooding. They also prioritize the

improvement of flood maps along with data modernization and the preparedness/education of citizens. Our findings could help modernize data because there are a lot of areas of Florida that haven't been updated in years and with the update of maps this could help get word out through the news, paper, or social media to help prepare communities and reach public areas that don't know that they are at a higher risk of flood than they do now. The OFM gets most of its budgeting from the state and agency appropriations, since this doesn't need a grant to do the research it should be easy for them to get the money to update the mapping for this research that we have done. Some barriers they may face is the mapping done may be hard to update or slow, by the time the new maps come out it may be too late by that point. Since we are including senior citizens, it may be harder to get the information out to them since they get their information different than the rest of the state. The barrier would be they may need printed maps instead of online or digital forms because they wouldn't understand the online versions or how to find them. Another barrier they may face could be the ability to reach out to each community that has senior citizens. Since the office is low staffed, it can be hard to get printed maps to every community in Florida to warn them about the upcoming flood risks.

Recommendation #1: To prioritize flood mitigation grants for high density senior communities in the spatial hotspots found. From our analysis, we can see that flooding affects senior citizens' communities. This would help the stakeholders because their goal is to help steer funding with that information and help the senior communities.

Recommendation #2: To improve data reporting and mapping in low-income senior communities where flood impacts may be underreported. Our analysis shows that flooding hits the senior citizen community and for the OFM, this would help make their data more accurate when looking at their floodplain maps.